Counting generators by their actual orbit requirements

Human Compression project: a component of the L019 recovery

Let $G = \langle \rho_0, \ldots, \rho_{r-1} \rangle$ be a transitive permutation group on Ω, with every maximal omission $G_j = \langle \rho_l : l \neq j \rangle$ intransitive. The generators are involutions. We will impose the string relations where they are needed. Suppose colour i has exactly one transposition $\{a, b\}$ joining different G_i-orbits. The acquired unique crossing lemma gives exactly two such orbits A, B, containing a, b respectively. Under the string relations, every ρ_j with $|j - i| > 1$ fixes a, b. All actions and orbit sets below are actual permutation actions; projected intersection properties are not assumed.

Lemma 1 (Minimal transitivity)**.** *Suppose a specified family of involutions has exactly c orbits on a set of d points. It contains a subfamily of at most $d - c$ members with the same orbits. If a subfamily of the marks of G_i is transitive on each of A, B, it contains every mark of G_i.*

Proof. Choose an inclusion-minimal subfamily with the same orbit partition. Deleting any member s splits one of its orbits; choose one transposition of s between two resulting orbits. The chosen edges form a forest, with parallel pairs counted as cycles. Indeed a cycle containing the chosen s-edge would connect its endpoints using only other chosen edges, hence using elements of the family with s deleted. This is impossible. The forest has at least c components and therefore at most $d - c$ edges.

For the second assertion, if ρ_j were absent from the subfamily, then G_j would contain that subfamily and ρ_i. The subfamily is transitive on both sides, and the crossing transposition joins them. Thus G_j would be transitive on Ω, a contradiction. □

Theorem 2 (Two imprimitive sides)**.** *If the actions of G_i on A, B preserve respectively $m_1, m_2 \geq 2$ blocks of sizes $k_1, k_2 \geq 2$, then*

$$r \leq m_1 + m_2 + k_1 + k_2 - 3 \leq (n + 2)/2, \qquad n = |\Omega|.$$

This assertion requires neither the string relations nor IP.

Proof. Choose a minimal subfamily M of the omission marks which is transitive on each of the two sets of blocks. Apply the first part of Lemma 1 to the simultaneous block action, which has exactly two orbits. It gives $|M| \leq m_1 + m_2 - 2$. This is minimality for transitivity, not a claim of independence or IP in the block image. Each point orbit of $\langle M \rangle$ meets every block in the corresponding side. Consequently the numbers c_1, c_2 of these point orbits satisfy $c_j \leq k_j$.

For each remaining omission mark ρ_l, choose an actual edge crossing two G_l-orbits. Such an edge exists because G_l is intransitive but G is transitive. Contract the $\langle M \rangle$-orbits. Its endpoints remain distinct, since $\langle M \rangle \leq G_l$. All the selected edges stay within A or within B. They form a forest: an alternative path round a cycle lifts through the $\langle M \rangle$-orbits to a path using no mark of colour l, contrary to its eligibility. Thus the $r - 1 - |M|$ remaining marks give at most $c_1 + c_2 - 2 \leq k_1 + k_2 - 2$ edges. This proves the first bound. For the second, expand

$$n + 2 - 2(m_1 + m_2 + k_1 + k_2 - 3) = (m_1 - 2)(k_1 - 2) + (m_2 - 2)(k_2 - 2) \geq 0.$$

□

Lemma 3 (Two commuting factors)**.** *Suppose the omission marks are partitioned into two nonempty subfamilies of sizes p, q, generating H, K respectively. Suppose H and K centralize one another and their product is transitive on B. Assume either A is a singleton or each of H, K is transitive on A. Let h be the common size of the H-orbits in B and let t be their number. Then*

$$p \le h-1, \qquad q \le t-1, \qquad h, t \ge 2, \qquad p+q \le |B|/2.$$

Proof. Because H and K commute, K permutes the H-orbits equivariantly. Their product is transitive on B, so K is transitive on this orbit set and transports each actual H-action to every other one. In particular all these orbits have size h, and $|B| = ht$.

Choose from the specified H marks a subfamily M_H that is transitive on one H-orbit, with $|M_H| \le h-1$ by Lemma 1. Equivariance makes it transitive on every H-orbit. Thus $\langle M_H, K\rangle$ is transitive on B. It is transitive on A too: A is either a singleton or K already acts transitively there. The second part of Lemma 1 forces every H mark into M_H, proving $p \le h-1$.

Choose a subfamily M_K of the specified K marks transitive on the t actual H-orbits, with $|M_K| \le t-1$. Then $\langle H, M_K\rangle$ is transitive on B and on A, using H on A when needed. Again minimality forces every K mark into M_K, so $q \le t-1$. Since p, q are positive, $h, t \ge 2$. Finally

$$p+q \le h+t-2 \le ht/2 = |B|/2,$$

where the middle inequality is $(h-2)(t-2) \ge 0$. No assertion of faithful factor restriction or quotient IP has entered the argument. □

Corollary 4 (A singleton and the two-point exception)**.** *Assume the string relations and $0 < i < r-1$. If $|A| = 1$, then $r \le (n+1)/2$. If $|A| = 2$ and both ρ_{i-1}, ρ_{i+1} act nontrivially on A, then $r \le n/2$.*

Proof. Take $H = G_{<i}$ and $K = G_{>i}$. They commute and have $p = i$, $q = r-i-1$ specified marks. In the first case apply Lemma 3 with singleton A and $|B| = n-1$. In the second both factors contain the transposition on A, so each is transitive there; apply the same lemma with $|B| = n-2$. In each case $r = p+q+1$ gives the claim. □

Theorem 5 (An interval side in the strict range)**.** *Assume the string relations, a connected noncommutation diagram, and $2r > n+2$. At a unique crossing, either the split is perfect after interchanging its two sides, or one side is primitive of degree at least two and its nontrivial distinguished restrictions occupy one interval adjacent to i. After a possible reversal, that side is B and $G_{<i}$ fixes B pointwise. If the split is not perfect, some label greater than i acts nontrivially on both sides.*

Proof. Both sides cannot be imprimitive by Theorem 2. A singleton side at an interior crossing is excluded by Corollary 4; at an endpoint crossing it immediately gives a perfect split, with the empty side of the index set assigned to the singleton. Thus we may choose a nontrivial primitive side, called B.

Suppose its active index set is disconnected. Partition it at a gap. The two generated images commute and are nontrivial normal subgroups of their primitive product. Each is therefore transitive. A nonidentity element of either factor centralizes the other transitive factor and has no fixed point. Hence every active distinguished restriction moves b. The unique crossing lemma confines its index to $i-1$ or $i+1$. There are exactly two active marks, one in each factor, and each alone is transitive. Thus $|B| = 2$ and both neighbours are active, contradicting Corollary 4 with the sides interchanged. Therefore the active set is one interval. It avoids i and contains a mark moving b, so it is adjacent to i. Reverse all indices if necessary to put it on the right; then $G_{<i}$ fixes B pointwise.

Suppose no mark is shared by the two sides. Every omission mark is nonidentity on exactly one side. On either interval of indices separated by i, this side cannot change at consecutive

positions: their supports would be disjoint, so those two marks would commute and disconnect the parent noncommutation diagram. Every right-hand mark consequently acts only on B, and every left-hand mark only on A. That is precisely a perfect split. Otherwise any shared mark belongs to the active interval of B, hence has index greater than i, proving the last assertion. □

Attribution and scope. The fracture framework and the both-imprimitive and primitive-side routes come from Cameron–Fernandes–Leemans, *The number of string C-groups of high rank*, arXiv:2212.12723v2, Section 5, and their earlier alternating-group source. The actual contracted orbit forest was already established in the accepted L044 work. Here minimal transitivity replaces the independent-group generation step; the two commuting-factor inequalities then give the singleton and two-point conclusions without evenness, faithful projected factors or projected IP. The primitive normal-subgroup argument is the acquired L044 argument, with its exact two-point boundary now settled by this count.

The only acquired mathematical input is the full unique-crossing and orbit-cut proof. All counting arguments are written above; no census, rank classification, computation or unproved stability statement is a premise. The remaining head support, split-existence and stability obligations are separate. No whole-original, conjecture, Lean or numerical-compression credit is asserted.